

Machine Learning for Early Detection of Cardiovascular Disease in Fiji

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Abstract— Cardiovascular disease (CVD) has long been a leading cause of premature death in Fiji, accounting for approximately 80% of annual fatalities. The consequential burden of these diseases has been significantly aggravated by their late detection, thereby influencing the substantial rise in fatalities and casualties. Presently, Fiji lacks research in machine learning-based predictive models that could potentially detect early signs and novel patterns of CVD, creating a critical need for innovative approaches in the field. To bridge this gap, this study presents a pioneering method utilizing ensemble machine learning to enhance CVD diagnosis in Fiji. This approach incorporates the k-means algorithm and five supervised machine learning algorithms: K-nearest neighbor (KNN), Decision tree, Random Forest, Naïve Bayes, and Support vector machine [SVM]. Subsequently, the precision of these models was evaluated using 10-fold cross-validation, aiming to identify the most efficacious machine learning model for CVD prediction. Based on the accuracy tests, the Random Forest model emerged as the most accurate among the four models, achieving an average accuracy of 89% and demonstrating its suitability for predicting CVD in Fiji.

Keywords—Machine learning, supervised learning, unsupervised learning, ensemble learning, cardiovascular disease prediction.

I. INTRODUCTION

CVD is a serious health concern in Fiji, a country ranked 25th worldwide for CVD-related annual deaths, according to the World Health Organization [1]. The late detection of CVD, often due to technological limitations, exacerbates health complications and strains healthcare resources. Fiji currently uses a long-term risk prediction method, which often results in late detection at a severe stage.

The objective of this research is to harness the predictive power of machine learning models, which would enable medical professionals to make more informed decisions regarding patient care, including the optimization of palliative supervision strategies and the rescheduling of patient appointments based on predicted results. The success of this approach depends on the timely input of accurate data into the prediction model. If accurate data is used for model training and prediction, there is a promising chance for early CVD detection, which can pave the way for the timely implementation of preventative measures. This could fundamentally transform the landscape of CVD management, promoting the health and longevity of Fijians.

This paper addresses late detection of CVD by proposing an early CVD detection system leveraging machine learning. The proposed method employs an ensemble of both unsupervised and supervised learning techniques, specifically the k-means algorithm and five supervised learning algorithms (KNN, Decision Tree, Random Forest, Naïve Bayes, and SVM). The accuracy of each supervised model is evaluated using 10-fold cross-validation. The methodology presented in this paper aims to reduce the health burden in Fiji by providing a more effective tool for early CVD detection.

II. LITERATURE REVIEW

Machine learning has revolutionized many industries, particularly medical healthcare systems, where it has been successfully used for disease prediction [2]. Researchers have highlighted its potential in managing various health conditions, including cardiovascular diseases, diabetes, liver disorders, tumors, and cancers [2]. Machine learning can also improve disease risk assessments. As a critical tool for healthcare professionals, it aids in disease detection at hospitals and significantly impacts healthcare decision-making processes, providing accurate advice that could potentially minimize casualties [3]. Thus, machine learning plays a crucial role in healthcare.

This study uses an ensemble machine learning method that combines both supervised and unsupervised learning models. Supervised learning, which builds models from labeled data, pairs each input with a corresponding output to make predictions for new data, often addressing classification problems [4]. In contrast, unsupervised learning creates models from unlabeled data, grouping similar data to discover new patterns and trends that can be helpful in classification tasks [4].

This research entailed an in-depth examination of the latest predictive models for heart attacks to understand previous methodologies, thereby laying a strong foundation for an efficient machine-learning solution for early detection of CVD. Decision tree algorithms were implemented in Clinical Decision Support Systems (CDSS) to assist in medical predictions of heart attacks [5]. The decision tree's performance was assessed via complexity analysis and proved to have higher accuracy than prior algorithms [5].

The Classification and Regression Trees (CS-CART) algorithm, known for its capacity to minimize test set errors, was utilized for heart attack prediction by applying a predefined set of rules to unclassified data [6]. The accuracy of the machine learning model was assessed through a 10-fold

cross-validation test, and a confusion matrix was built to denote the diverse possible outcomes [6]. This facilitated the evaluation of the model's sensitivity, specificity, and overall accuracy. Among the tested models, the CS-CART algorithm exhibited the highest accuracy, indicating its superior performance.

A research study evaluated five supervised machine learning algorithms using two dimensionality reduction techniques—feature extraction and feature selection—to predict heart disease [7]. The Principal Component Analysis (PCA) was used for feature extraction. Five machine learning algorithms were then applied to the dataset, and their accuracies were measured using a confusion matrix. The results were as follows: Naïve Bayes showed 83.49% accuracy, Support Vector Machine achieved 85.76%, KNN scored 83.16% with K=3, the Decision Tree reached 92.2%, and the Random Forest had the highest accuracy of 97.7% [7].

The study by Yadav, Gediya, & Kazi [8] examined the use of Support Vector Machines (SVM) and other supervised learning algorithms, including KNN, Decision Tree, K-Means, and Naïve-Bayes, for predicting CVD [8]. The researchers developed a machine learning system using a dataset of 70,000 medical records, which was split into training and testing sets, encoded, and scaled during pre-processing. After applying the selected machine learning algorithms, a confusion matrix was used to evaluate their performance. The KNN algorithm exhibited an accuracy rate of 63.4% [8]. Despite its high processing time, the KNN model was found to be less susceptible to noise data, suggesting its potential for use in large-scale applications [8].

The research conducted by Jindal [9] demonstrated that the KNN algorithm outperformed both logistic regression and random forest algorithms in predicting cardiovascular disease. Using the Cleveland dataset, which consists of 303 instances and 14 attributes, a cardiovascular prediction system was developed. The process began with feature selection for dimensionality reduction, after which the data was split into training and testing sets. According to the confusion matrix used to assess model performance, the KNN algorithm yielded an accuracy of 88.52%, superior to the 85% accuracy of the random forest algorithm and the 83% accuracy of the logistic regression algorithm [9].

III. METHODOLOGY

To achieve the research objective, a systematic approach was implemented, which was divided into five distinct stages: data collection, data pre-processing (unsupervised learning), application of unsupervised machine learning (KNN), data pre-processing (supervised learning), and subsequent use of supervised machine learning techniques, followed by the evaluation of the supervised machine learning model. Figure 1 shows a flowchart devised to fulfil the objective.

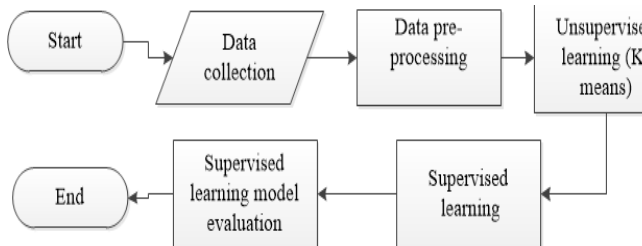


Figure 1: Flowchart of methodology

A. Data collection

Various online datasets are available for training and testing CVD risk prediction models. Two primary considerations guided the selection of the best datasets: the similarity of contributing risk factors to those identified by Fiji's Ministry of Health and Medical Services and the datasets' quality and volume to assess the model's scalability. Scalability refers to the model's ability to effectively process large amounts of new data while maintaining consistent results and managing computational complexity.

To test the machine learning model's capabilities, six datasets were selected: the Heart Failure Prediction [10], Cleveland [11], and Hungarian, Switzerland, Long Beach VA, a comprehensive CVD dataset [12]. These datasets were obtained from Kaggle and the University of California, Irvine's (UCI) website, allowing robust testing of the model across multiple data volumes and distribution changes. Each of the six datasets contains 12 attributes. Of these, 11 attributes serve as the risk factors contributing to heart disease, while the 12th attribute acts as the target class or label. This label indicates whether heart disease is present or absent. Table 1 below shows a detailed description of each attribute.

TABLE I. ATTRIBUTE DESCRIPTION

Attribute	Description
Age	States patient's age in years ranging from 28 to 77.
ChestpainType	Describes chest pain of 4 types ASY, ATA, NAP, TA.
RestingBP	Resting blood pressure measured in mm Hg when a patient has been admitted ranges from 0 to 200 mmHg.
Cholesterol	Cholesterol measured in mm/dl ranging from 0 to 603 mg/dl.
FastingBS	Fasting blood sugar measured in mg/dl where: 1 = true indicates the presence of diabetes 0 = false indicates the absence of diabetes.
RestingECG	Resting electrocardiographic results normal = normal ECG; ST = abnormality in ECG; LVH = indicates probable or definite left ventricular hypertrophy.
MaxHR	Describes a heart rate.
ExerciseAngina	Indicates exercise-induced angina, Y = yes and N = no.
Oldpeak	ST depression induced by exercise relative to rest.
ST_Slope	The peak exercise ST segment.
HeartDisease	Target class determining cardiovascular disease. 1 = present 0 = absent.

B. Data pre-processing

This study relied heavily on data pre-processing to ensure the datasets were suitable for the predictive model specific to

Fiji's needs. The aim was to adjust the data attributes to correspond with Fiji's prevalent cardiovascular risk factors and to address any errors or inconsistencies within the data acquired from online sources.

In Fiji, healthcare professionals utilize the third edition of cardiovascular therapeutic guidelines, introduced in 2015 by the Ministry of Health and Medical Services and based on Australian standards, to assess cardiovascular risk [13]. These guidelines provide a comprehensive approach to cardiovascular management, which includes risk assessment, identifying new lifestyle-related risk factors, and advice on treatments and referrals. The initial data required modifications to suit the Fijian context. Unnecessary attributes were eliminated, and cholesterol data were converted from mg/dl to mmol/l to conform to the format used in Fiji. These initial steps were crucial for shaping a reliable model suitable for the machine learning tool used in Fiji. The six tasks related to data pre-processing included:

1. Removing irrelevant attributes - irrelevant attributes from the online sample datasets were removed based on the risk factors outlined in Fiji's cardiovascular therapeutic guidelines. Irrelevant attributes were removed, including ChestPainType, RestingECG, ExerciseAngina, Oldpeak, MaxHR, ST_Slope, and HeartDisease, leaving only systolic blood pressure, cholesterol levels, smoking status, age, gender, and diabetic status.
2. Converting cholesterol attribute - The cholesterol measurements in all the datasets had to be adjusted to the format used in Fiji. As per the cardiovascular therapeutic guidelines, Fiji gauges cholesterol levels in millimoles per litre (mmol/l), a standard metric system unit. However, the datasets initially presented cholesterol levels in milligrams per deciliter (mg/dl). Therefore, all cholesterol values in the datasets were converted to the mmol/l format.
3. Encoding categorical data - Failing to convert categorical data would affect the model's prediction accuracy and increase computational complexity [14]. Categorical data was encoded for better compatibility with machine learning models. Label encoding was applied to the Sex attribute.
4. Removing duplicates - Duplicate instances, which could cause overfitting, were removed from the datasets.
5. Removing outliers - Outliers, or data points far from the typical data distribution, can stem from errors in data entry or inaccuracies in CVD diagnosis [15]. Outliers were addressed using the Z-score technique. Any record with a Z-score beyond three standard deviations from the mean was considered an outlier and removed.
6. Removing missing values - Missing values were identified and removed to avoid biases in CVD prediction. These were represented by 0's or empty cells in the datasets. All records with missing values were removed to ensure the reliability of model performance.

C. Unsupervised learning (K-means)

This research intends to create a machine-learning methodology predicting five different CVD risk levels, from risk level one to risk level five. However, the datasets currently used for model development and testing lack a specific class or target feature indicating each record's CVD risk level. Consequently, these datasets can't be used directly in their current form to develop the machine-learning model. The K-means algorithm, an unsupervised learning method, is the most appropriate solution for this problem. The K-means algorithm segregates data into distinct clusters, each with a centroid, thereby grouping similar data together [16].

This study used 2,000 records from the online dataset repository to test and validate the K-means model. This manageable sample size helped to run computations efficiently, especially with iterative algorithms like K-means. This balance between sample size and feasibility of analysis suited the limited time for the study, providing enough data for reliable interpretation without overwhelming our analysis.

Before developing the K-means model, the initial step involved splitting the primary datasets into sixty-four smaller datasets. The segmentation followed a specific strategy, each subset representing a unique set of CVD cases across different risk levels. The categorization was determined by factors such as the individuals' ages, their smoking habits, whether they are diabetic or non-diabetic, and their systolic blood pressure and cholesterol readings. This approach offered an in-depth understanding of the potential risk levels associated with each specific group, thereby enabling the K-means model to cluster data efficiently per category.

In this study, an advanced form of the K-means algorithm, the K-means++, was implemented. This variant improves the selection of initial centroids spread evenly across the data space, thereby mitigating the risk of inadequate clustering resulting from random centroid initialization [17]. The K-means++ method initially picks a random data point as the first centroid and then computes the Euclidean distance between the remaining data points and the closest centroid [18]. The selection of subsequent centroids is driven by a probability directly proportional to the square of their distance from the nearest current centroid [19]. This approach facilitates the selection of optimal centroids, with distant data points more likely to be chosen.

This iterative process continues until the desired number of centroids is achieved. Upon determining the best centroids, the standard K-means algorithm then groups the data based on the centroids identified by the K-means++ technique. A total of 64 distinct K-means models were developed during this research, each custom-fitted to a unique dataset within a specific category that defined age, gender, smoking, and diabetic status. After building 64 K-means models, the clusters created by each model were classified. The K-means algorithm divided the data into groups based on similarities and were labelled risk level 0 to 5. Afterwards, these groups were given clear names based on their unique CVD characteristics in each category. The K-means algorithms transformed the unlabeled dataset into a labelled dataset, which is helpful for training models in supervised learning.

D. Supervised learning

In this research, six supervised machine-learning models were used to analyze data clusters generated by the K-means models. These models were built using Python's Scikit-learn

(Sklearn), a powerful library with many machine-learning classes and tools. When developing the models, the parameters of the models were considered. Parameters play a crucial role in aiding the model in learning data patterns as they shape the model and significantly influence the quality of the model's outcomes.

One of the models utilized was the KNN algorithm, known for its simplicity and effectiveness in predicting CVD risk. The KNN model determines new data classification by identifying and analyzing the K nearest data points or neighbors [20]. Two critical parameters guided our use of the KNN model in this study: the number of neighbors and the distance metric. It was observed that the selection of neighbors is data-dependent and can heavily influence classification accuracy. We used both Euclidean and Manhattan methods for distance metrics, emphasizing that the choice of metric should consider the importance of individual CVD risk factors.

The Decision Trees were also used for CVD prediction. This model creates a tree-like structure of the training dataset, dividing it into branches until each reaches a 'leaf' node - the prediction outcome [21]. A key parameter in this model is the criterion, which measures the efficiency of data partitioning. Our study explored the criterion parameter using two methods - gain ratio and Gini impurity.

The Random Forest was another model employed in this study. This model constructs numerous decision trees by randomly creating subsets of the original medical dataset in a process called bootstrapping [22]. The model then collates the results of all trees through a voting mechanism to formulate the final CVD risk prediction model. This study primarily focused on two parameters to shape Random Forest models: the number of trees and the criterion. It was found that increasing the number of trees enhances the predictive accuracy, and the criterion parameter (explored using gain ratio and Gini impurity methods) determines the efficiency of data partitioning.

In addition, the Naïve Bayes algorithm was also used for predicting CVD risk. The algorithm works on the principles of Naïve and Bayes, assuming independence between each CVD risk factor and altering the estimated likelihood of a specific CVD risk when new information is introduced [23]. A variant of Naïve Bayes, the Multinomial Naïve Bayes (MNB), was tested in this study. The MNB model calculates the frequency of each CVD risk factor within the training data and uses these to determine the probability of each factor for the given CVD risk level.

Lastly, the SVM model was also utilized in our research. The SVM model predicts CVD risk levels by finding an optimal hyperplane that maximizes the distance between it and the nearest data points from each CVD risk level [24]. This complex but powerful process transforms CVD risk factors into a higher-dimensional space to enable easier separation. A significant parameter in the SVM model is the kernel function; in this study the polynomial kernel was utilized, which facilitates handling both linearly and non-linearly separable data.

E. Model Evaluation

Evaluating the effectiveness of the suggested methodology is essential because of the serious repercussions that could arise from incorrect forecasts, such the possibility of misdiagnosis that could have disastrous effects.

Consequently, in order to guarantee that the approach produced accurate predictions, it was imperative that it be thoroughly verified. A subset of 500 records were initially assessed, and that number was gradually increased to 2000. This offered a starting point for performance benchmarking. At every phase, we methodically added 500 more records to the collection. This iterative approach promoted training process robustness and enabled a thorough analysis of each model's capacity to adjust to variations in dataset size and distribution. Four distinct and ever bigger subsets were thus produced, leading to the creation of an extensive dataset with 2000 records. This gave the five machine learning models a variety of data to work with during their learning phase, allowing for a comprehensive and exacting assessment of their performance.

To replicate the unpredictable and diverse kind of data that machine learning models are likely to encounter in real-world scenarios—specifically, in Fiji—the dataset was partitioned into 2000 records. To evaluate each model's resilience to randomness and shifts in data distribution, 500 records were added incrementally at each stage. This allowed us to determine which model was the most stable and dependable for use in practical settings. Preliminary analysis suggested that extending the dataset over 2000 records did not yield appreciable gains in model performance, therefore we opted to stop here. This plateau indicated that our models had reached a steady state of prediction accuracy, therefore 2000 records were deemed a sufficient size for a thorough analysis. As a result, this strategy provided a strict process for creating a model that was appropriate for the difficulties posed by actual situations in Fiji.

As part of the evaluation, a 10-fold cross-validation testing procedure was implemented for each dataset to ensure a comprehensive evaluation of the models. The cross-validation process involves dividing the entire dataset into K distinct subsets or folds. Each model was trained on the remaining folds and then tested one-fold at a time, allowing each model to be trained and evaluated ten times on different dataset sections [25]. This technique utilizes the entire dataset for training and testing, providing a reliable performance estimation and minimizing the risk of overfitting the data.

When evaluating the risk levels of CVD using a multiclass prediction model, accuracy is a more reliable and complete indicator than sensitivity and specificity [26]. Accuracy captures the model's performance across all risk categories, from low to high, in contrast to sensitivity and specificity, which are primarily designed for binary classification problems and concentrate on the model's performance relative to a particular positive class. This is especially important as each risk category has different clinical consequences and treatment options, thus generating accurate predictions for each is essential. We are able to obtain a comprehensive understanding of the model's predictive capacity over the whole range of CVD risk levels by using accuracy as the performance parameter. Given an equally distributed or equally relevant dataset, this single, unified metric offers an intuitively comprehensible measure that naturally balances the performance across all classes. Thus, in situations when each risk category has unique clinical importance, accuracy is a more relevant and all-encompassing metric to assess the overall performance of the model.

IV. MODEL EVALUATION RESULTS

Every model in this study was subjected to rigorous testing using each of the four datasets. To identify the most successful model, the average accuracy achieved on each dataset was evaluated. This approach allowed us to consider how each model performed across various data, thereby preventing the selection of a model that only performed well on one specific dataset. The model with the highest average accuracy across all four datasets was ultimately identified as the top-performing model.

The testing phase, which involved 500 records and five unique machine-learning models, produced fascinating outcomes, as depicted in Table 2. The KNN model showed a robust performance with an average accuracy of 75%, attributed to its proficiency in handling smaller datasets. The model used two neighbors and the Euclidean distance metric. However, the Random Forest demonstrated an average accuracy of 86%. It used the Classification and Regression tree and created 100 decision trees. The Naive Bayes model, specifically the Gaussian Naive Bayes, managed to secure an average accuracy of 75%, potentially due to some independent features in the dataset. The SVM model also achieved a 75% average accuracy. Lastly, the Decision Tree model achieved an average accuracy of 86% using the Classification and Regression tree.

TABLE II. RESULTS OF MODEL ACCURACY BASED ON A DATASET OF 500 RECORDS

Model	Accuracy
KNN	75.3 %
Random Forrest	86%
Naïve Bayes	75.6%
SVM	75%
Decision tree	86%

During the subsequent testing phase with a larger dataset of 1000 records, there were slight alterations in the model's accuracy, which can be seen in Table 3. The KNN model improved from 75% to 78% with an increase in dataset size, thanks to more reliable majority voting with three neighbors. The Random Forest model's accuracy also rose from 86% to 90% due to the model's capacity to handle larger datasets effectively. However, the Naive Bayes model's accuracy fell from 75% to 73%, potentially due to the impact of larger data on its feature independence assumption. The SVM model's accuracy dropped from 75% to 69%, while the Decision Tree model saw a slight boost from 86% to 87%, possibly because larger datasets allowed it to identify more intricate data patterns.

TABLE III. RESULTS OF MODEL ACCURACY BASED ON A DATASET OF 1,000 RECORDS

Model	Accuracy
KNN	78.4 %
Random Forrest	90%
Naïve Bayes	73.2%
SVM	69%

Decision tree	87%
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During the subsequent testing stage, the machine learning models were subjected to an expanded dataset, this time comprising 1500 records, as indicated in Table 4. The KNN model saw improvement, with its accuracy increasing from 78% to 83%, likely due to a more diverse nearest neighbor set in the larger dataset. The Random Forest model's accuracy improved slightly from 90% to 91%, showing its ability to leverage more data effectively. In contrast, the Naive Bayes model's accuracy dropped slightly from 73% to 72%. The SVM model's accuracy also decreased from 69% to 66%, possibly due to difficulty handling high-dimensional data or specific data distributions. The Decision Tree model, however, improved slightly from 87% to 89%. Despite variations, the Random Forest model continued to outperform others with an accuracy of 91%, showcasing the strength of ensemble learning methods with larger datasets.

TABLE IV. RESULTS OF MODEL ACCURACY BASED ON A DATASET OF 1,500 RECORDS

Model	Accuracy
KNN	83 %
Random Forrest	91.1%
Naïve Bayes	72%
SVM	66%
Decision tree	89%

In the concluding testing stage involving the most extensive dataset of 2000 records, several variations in the performance of the models were noticed, as depicted in Table 5. The KNN model continued to improve, with its average accuracy rising from 83% to 84%. Conversely, the Random Forest model showed a slight performance decrease, with its average accuracy dropping from 91% to 89%. The Naive Bayes model also saw a minor decrease in average accuracy from 72% to 71%. On the other hand, the SVM model showed improvement, with its accuracy rising from 66% to 72%, possibly due to better performance with high dimensional data. Finally, the Decision Tree model's performance decreased slightly, dropping its average accuracy from 89% to 90%.

TABLE V. RESULTS OF MODEL ACCURACY BASED ON A DATASET OF 2,000 RECORDS

Model	Accuracy
KNN	84 %
Random Forrest	89%
Naïve Bayes	71%
Decision tree	90%

V. CONCLUSION

Our extensive set of tests concludes that the CART Random Forest model is the most appropriate for estimating the risk of CVD in Fiji. This model outperformed other machine learning models in terms of accuracy over a wide range of dataset sizes. Notably, the CART model can handle big, complicated datasets with ease and is resistant to overfitting. It also provides a feature importance metric. Our approach, when combined with the ensemble learning

methodology and K-means algorithm, represents a substantial leap in predictive analysis relevant to Fiji's existing CVD landscape by providing a novel manner of detecting emergent patterns in CVD risk.

It is important to recognize certain limitations. Python was primarily used to develop the CART Random Forest model; nevertheless, end users may find Python difficult to use or comprehend in a real-world setting. As a result, direct implementation of the model is difficult in its current form because it assumes some knowledge of Python programming from the user. In order to solve this problem, future research will concentrate on creating a system that is client-server based. This would allow for a more user-friendly interface—possibly in the form of a mobile application—that would make it easier for users to estimate their risk of CVD, while the model itself would be hosted on a server for greater accessibility to a wider audience.

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